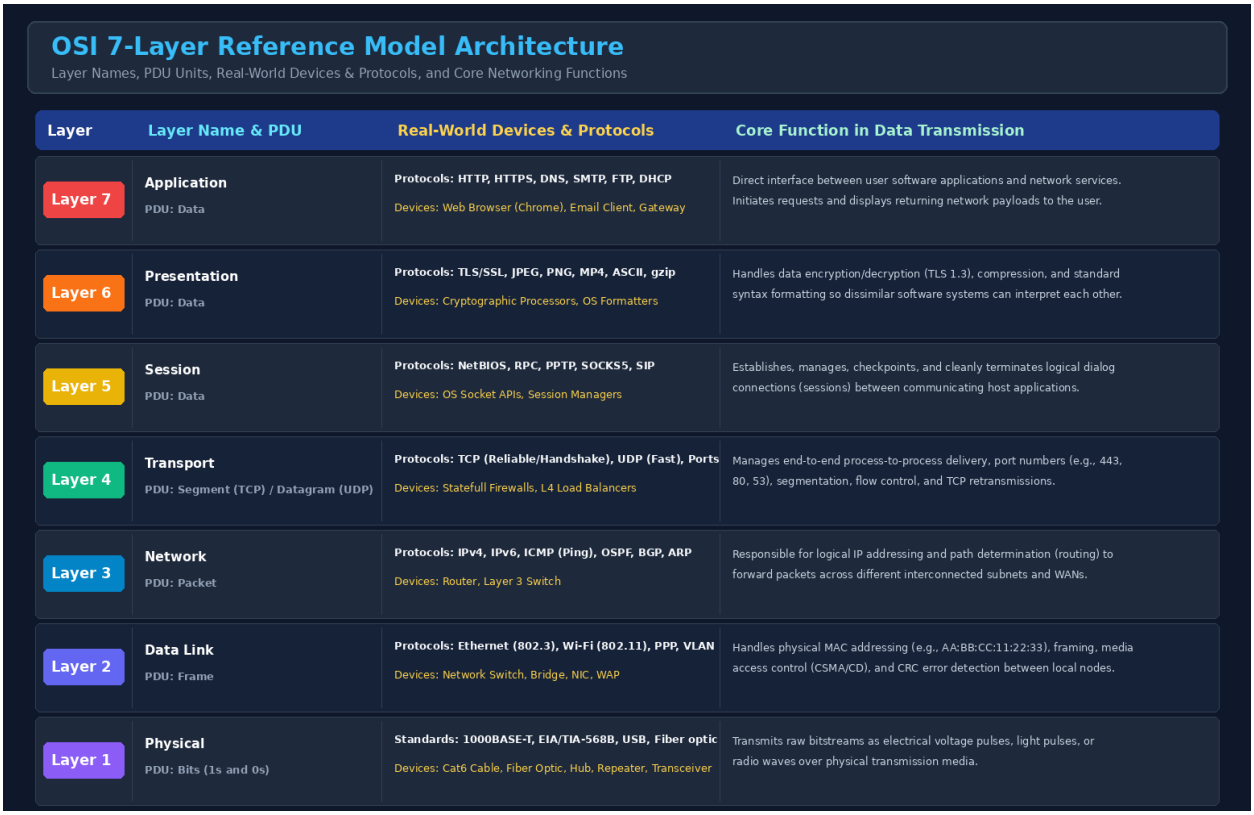
OSI Layer

1. OSI 7-Layer Model Diagram & Reference Table



2. Scenario Matching

- **a) Encrypting data before sending:**

$\rightarrow$ **Presentation Layer (Layer 6)**

(Handles data encryption, decryption, and secure TLS/SSL cipher formatting).

- **b) Routing packets across networks:**

$\rightarrow$ **Network Layer (Layer 3)**

(Uses logical IP addressing and routing tables to determine the best path between different subnets).

- **c) Converting data to electrical signals:**

$\rightarrow$ **Physical Layer (Layer 1)**

(Translates binary bits into physical electrical voltage pulses, light pulses, or RF radio waves).

- **d) Opening a web page in your browser:**

$\rightarrow$ **Application Layer (Layer 7)**

(Initiates HTTP/HTTPS GET requests directly via user-facing browser software).

- **e) Detecting and correcting errors in frames:**

$\rightarrow$ **Data Link Layer (Layer 2)**

(Encapsulates data into frames and uses Cyclic Redundancy Checks [CRC] and Frame Check Sequences [FCS] to identify transmission errors).

3. OSI Layers in Action: WhatsApp Messaging

When you send a message (e.g., *"Hello!"*) on **WhatsApp**, multiple OSI layers handle the transmission:

1. **Layer 7 (Application Layer):**

The WhatsApp client interface accepts your input, formats the text message payload, and calls application messaging APIs to trigger the send action.

2. **Layer 6 (Presentation Layer):**

WhatsApp applies **End-to-End Encryption (E2EE)** using the Signal Protocol, converting your plain text into unreadable ciphertext before it leaves your device so only the intended recipient can decrypt it.

3. **Layer 4 (Transport Layer):**

WhatsApp segments the encrypted message and uses **TCP** (typically over port 443 or 5222) to establish a reliable, sequenced connection, guaranteeing delivery through acknowledgments and automatic retransmissions if packets drop.

4. **Layer 3 (Network Layer):**

Attaches your phone's local IP address (source) and WhatsApp's server IP address (destination), encapsulating the transport segment into an IP packet for routing across the internet.

4. Scenario Diagnosis: Can Ping IP, but Cannot Access Website

- **Identified Failing OSI Layers: Layer 7 (Application Layer)** and/or **Layer 4 (Transport Layer)**.

Technical Reasoning:

- **Why Layers 1, 2, and 3 are working correctly:**

A successful ping uses the **ICMP (Internet Control Message Protocol)** operating at **Layer 3 (Network Layer)** across your physical cables (Layer 1) and local switch/Wi-Fi (Layer 2). Because ping replies return successfully, the physical media, MAC frame delivery, IP routing, and subnet gateways are fully operational.

- **Why the failure lies at the higher layers:**

    1. **Layer 7 (DNS Resolution Failure):** If you are accessing the site via a domain name (e.g., www.example.com), the DNS server might have failed to resolve the name to its corresponding IP address.
    2. **Layer 7 (Web Server / HTTP/HTTPS Service Down):** The remote web server software (such as Nginx, Apache, or Cloudflare) might have crashed or returned an HTTP 500 error, even though the host operating system's network stack responds to basic ICMP pings.
    3. **Layer 4 (Port Blocking / Firewall Filtering):** Network firewalls or security groups might permit ICMP echo traffic (Ping) while actively filtering out or dropping TCP traffic on **Port 80 (HTTP)** and **Port 443 (HTTPS)**.